THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 5 & 6: **Describing Motion Around Us, How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A cyclist covers 900 m in 50 s along a straight road without turning back. The average speed is: 1

(a) 0.06 m s^{-1} **(b)** 18 m s^{-1} **(c)** 36 m s^{-1} **(d)** 9 m s^{-1}

2. A speed of 108 km h^{-1} in SI units is: 1

(a) 60 m s^{-1} **(b)** 388.8 m s^{-1} **(c)** 36 m s^{-1} **(d)** 30 m s^{-1}

3. The velocity of a car changes from 15 m s^{-1} to 0 m s^{-1} in 5 s in a straight line. Its average acceleration is: 1

(a) −6 m s^{-2} **(b)** −3 m s^{-2} **(c)** 3 m s^{-2} **(d)** 4 m s^{-2}

4. A body starts with velocity 0 m s^{-1} and accelerates uniformly at 2 m s^{-2} for 5 s. The distance travelled is: 1

(a) 10 m **(b)** 12.5 m **(c)** 25 m **(d)** 50 m

5. A vehicle moving at 30 m s^{-1} is brought to rest with a uniform retardation of 5 m s^{-2}. The stopping distance is: 1

(a) 180 m **(b)** 90 m **(c)** 45 m **(d)** 6 m

6. The net force needed to give a body of mass 2 kg an acceleration of 6 m s^{-2} is: 1

(a) 24 N **(b)** 8 N **(c)** 12 N **(d)** 0.33 N

7. A net force of 100 N acts on a body of mass 25 kg. Its acceleration is: 1

(a) 2500 m s^{-2} **(b)** 8 m s^{-2} **(c)** 0.25 m s^{-2} **(d)** 4 m s^{-2}

8. The weight of a 60 kg body is (g = 10 m s^{-2}): 1

(a) 300 N **(b)** 600 N **(c)** 60 N **(d)** 1200 N

9. Two forces of 15 N and 9 N act on a block along the same straight line in opposite directions. The magnitude of the net force is: 1

(a) 24 N **(b)** 135 N **(c)** 6 N **(d)** 15 N

10. A body of mass 5 kg initially at rest is pushed by a net force of 10 N for 3 s. Its final velocity is: 1

(a) 30 m s^{-1} **(b)** 3 m s^{-1} **(c)** 12 m s^{-1} **(d)** 6 m s^{-1}

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What does the slope of a position–time graph represent? 2

12. What does the area under a velocity–time graph represent? How is it found for constant velocity? 2

13. Why is it difficult to walk on a wet polished floor? 2

14. A box is pushed with a force equal to the friction acting on it. Will it move with constant velocity? Why? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Derive $s = ut + \frac{1}{2}at^2$ from a velocity–time graph. 3

16. A 0.1 kg bullet is fired from a 5 kg gun with a force of 2 N. Find the initial accelerations of the bullet and gun. 3

17. A train starting from rest reaches 20 m s^{-1} in 50 s. Find the acceleration and distance. 3

OR

A car moving at 20 m s^{-1} brakes with retardation 5 m s^{-2}. Find the stopping distance and time. 3

18. Why does the Earth not appear to move towards a falling fruit although the forces are equal? 3

OR

Explain the working of a balloon rocket. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A driver at 20 m s^{-1} sees an obstacle 50 m ahead and brakes at 5 m s^{-2}. 5

(i) Find the stopping distance.
(ii) Will the vehicle hit the obstacle?
(iii) If speed were 30 m s^{-1}, find the stopping distance.
(iv) What do you infer about speed limits?

OR

(i) Define uniform circular motion. 5
(ii) Why is it accelerated?
(iii) Show with a marble in a ring that velocity is along the tangent.
(iv) State why it is an idealised model.

20. (i) What is friction? 5
(ii) Describe the experiment to compare friction on different surfaces.
(iii) How does friction help us to walk?

OR

Case study: A car of 800 kg moving at 25 m s^{-1} stops in 10 s. 5
(i) Find the retardation.
(ii) Find the braking force.
(iii) Find the stopping distance.
(iv) If the speed were doubled for the same force, find the new stopping distance.

END OF QUESTION PAPER — **Total: 40 Marks**